Lv2 ポリプ

中3数学　3章　二次方程式　／　基礎問題20問

(1) $8x^2=288$

答え　$\boldsymbol{x=\pm 6}$

$x^2=36$

(2) $3x^2=1$

答え　$\boldsymbol{x=\pm\dfrac{\sqrt{3}}{3}}$

$x^2=\dfrac{1}{3}$、$x=\pm\dfrac{1}{\sqrt{3}}=\pm\dfrac{\sqrt{3}}{3}$

(3) $3x^2=\dfrac{4}{3}$

答え　$\boldsymbol{x=\pm\dfrac{2}{3}}$

$x^2=\dfrac{4}{9}$

(4) $(x+5)^2=49$

答え　$\boldsymbol{x=2,\ -12}$

$x+5=\pm 7$

(5) $(x-8)^2=3$

答え　$\boldsymbol{x=8\pm\sqrt{3}}$

$x-8=\pm\sqrt{3}$

(6) $80-(x-7)^2=0$

答え　$\boldsymbol{x=7\pm 4\sqrt{5}}$

$(x-7)^2=80$、$x-7=\pm 4\sqrt{5}$

(7) $x^2+10x+24=0$

答え　$\boldsymbol{x=-4,\ -6}$

$x^2+10x+5^2=-24+5^2$

$(x+5)^2=1$、$x+5=\pm 1$

(8) $x^2-8x-32=0$

答え　$\boldsymbol{x=4\pm 4\sqrt{3}}$

$x^2-8x+4^2=32+4^2$

$(x-4)^2=48$、$x-4=\pm 4\sqrt{3}$

(9) $x^2+5x-24=0$

答え　$\boldsymbol{x=3,\ -8}$

$x=\dfrac{-5\pm\sqrt{25+96}}{2}=\dfrac{-5\pm 11}{2}$

(10) $3x^2+3x-2=0$

答え　$\boldsymbol{x=\dfrac{-3\pm\sqrt{33}}{6}}$

$x=\dfrac{-3\pm\sqrt{9+24}}{6}$

(11) $x^2+5x+2=0$

答え　$\boldsymbol{x=\dfrac{-5\pm\sqrt{17}}{2}}$

$x=\dfrac{-5\pm\sqrt{25-8}}{2}$

(12) $2x^2-6x-5=0$

答え　$\boldsymbol{x=\dfrac{3\pm\sqrt{19}}{2}}$

$x=\dfrac{6\pm\sqrt{76}}{4}=\dfrac{6\pm 2\sqrt{19}}{4}$ → 2で約分

(13) $5x^2-2x-2=0$

答え　$\boldsymbol{x=\dfrac{1\pm\sqrt{11}}{5}}$

$x=\dfrac{2\pm\sqrt{44}}{10}=\dfrac{2\pm 2\sqrt{11}}{10}$ → 2で約分

(14) $3x^2-5x+2=0$

答え　$\boldsymbol{x=1,\ \dfrac{2}{3}}$

$x=\dfrac{5\pm\sqrt{25-24}}{6}=\dfrac{5\pm 1}{6}$

(15) $x^2-7x+10=0$

答え　$\boldsymbol{x=2,\ 5}$

$(x-2)(x-5)=0$

(16) $x^2-x-42=0$

答え　$\boldsymbol{x=-6,\ 7}$

$(x+6)(x-7)=0$

(17) $5x^2=6x$

答え　$\boldsymbol{x=0,\ \frac{6}{5}}$

$5x^2-6x=0$、$x(5x-6)=0$

(18) $x^2-14x+49=0$

答え　$\boldsymbol{x=7}$

$(x-7)^2=0$

(19) $36-x^2=0$

答え　$\boldsymbol{x=\pm 6}$

$x^2=36$

(20) $x(x+5)=-4$

答え　$\boldsymbol{x=-1,\ -4}$

$x^2+5x+4=0$、$(x+1)(x+4)=0$